

The Metamorphosis of Storytelling: Time-based Interactivity in Virtual Reality Filmmaking

Kaifeng Wu

Yale University Center for Collaborative Arts and Media
New Haven, USA
kaifeng.wu@yale.edu

Dana Karwas

Yale University Center for Collaborative Arts and Media
New Haven, USA
dana.karwas@yale.edu

ABSTRACT

Our work addresses a novel technique for free navigation in a Virtual Reality film and explores the spatial - temporal narrative implications of such a technique.

With the increasing popularization of VR filmmaking, under current technology, filmmakers who choose to represent a realistic scene are faced with a choice between either graphically intensive real-time 3D rendering on a high-end VR system, or a solution based on Omnidirectional Stereoscopic (ODS) Imaging captured by a camera. Commonly known as 360 videos, they suffer from a fixed viewpoint. To address this specific issue, many recent advancements in the field proposed novel view reconstruction from spherical panoramas, planar image - based rendering, light field rendering, etc. for free-viewpoint navigation.

We explore the interactive implications of implementing such techniques for VR filmmaking. Specifically, the temporal relationship of viewpoints and the experience of navigating through them. From an initial proof of concept with a linear path, we continued to develop the technique through installations and experimented with an infinitely looping circular path around NYC grand central terminal's clock tower. We aim to integrate multiple optimizations and techniques into an easy-to-use open-source package for filmmakers. Including storage optimizations, novel view synthesis, and locomotion redirection.

We have found that while the technical package is complicated to explain, the experience is delightfully intuitive. By sharing our discoveries with the world, we hope to enable filmmakers of the future to tell compelling stories that matter.

CCS CONCEPTS

• **Computing methodologies** → **Computer graphics**; *Computer vision*.

ACM Reference Format:

Kaifeng Wu and Dana Karwas. 2024. The Metamorphosis of Storytelling: Time-based Interactivity in Virtual Reality Filmmaking. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '24)*, May 11–16, 2024, Honolulu, HI, USA. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3613905.3648672>

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).
CHI EA '24, May 11–16, 2024, Honolulu, HI, USA
© 2024 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-0331-7/24/05
<https://doi.org/10.1145/3613905.3648672>

1 INTRODUCTION

Drawing upon our interest in science fiction time machines and visual perception research, this paper explores the transformative impact of locomotion on narrative construction in Virtual Reality (VR) filmmaking. We introduce a pioneering system designed to challenge traditional locomotion methods in immersive Omnidirectional Stereoscopic Environments (ODS), establishing novel connections between memory, the human body, and virtual space. The system presents a dichotomy between two elements: spatio-temporal filming and non-sequential playback techniques. This is demonstrated through our Pacing Time immersive installation that serves as the primary case study for this paper. In the filming of Pacing Time, we developed a system that records the precise spatial location of the camera for each frame. This affords us the opportunity to introduce a playback that diverges from traditional sequential playback; instead, our system dynamically presents frames based on the real-time tracked position of the viewer's headset as they navigate through the scene. Our goal is to generate possibilities and potential implications of implementing such time-based techniques in standardized tools for future VR filmmakers.

Unique to each form of storytelling, we believe there is a defining technique that elevates it towards a form of art - a story experience only VR film can deliver. Drawing parallels with Eisenstein's theory of Montage in cinema development, Pacing Time unlocks a new way to tell a story across space and time that's unique to VR filmmaking. Pacing Time showcases the feasibility of occupying a specific point in space functioning both as a Point of View (POV), and a non-sequential, non-irreversible time array. In addition to bringing up philosophical questions around what is reality?, and cognitive inquiries into what is memory? We find this installation and the system driving it to be significant as it presents multiple opportunities for the VR filmmakers and audiences. From the development standpoint, VR filmmakers can seamlessly integrate this temporal control into content creation. This presents interesting challenges for communal viewing experience, as each participant dictates their own temporal trajectory. By introducing degrees of freedom in time control within a scene, the user experience becomes entirely customizable, allowing users to dictate the timing and progression of the VR film.

2 NARRATIVE

The filmmaking medium has always been interconnected with the advancement of technology. Each technological push, from the inception of the cinematograph in 1895 to the digital filmmaking era, each technological leap continuously shaped and evolved the film medium. Technical innovations have equipped filmmakers powerful, profoundly impacting how narratives are crafted and presented.

VR filmmaking represents a significant leap in this dynamic, introducing the possibility of interactivity. This shift from passive to active audience participation has fundamentally transformed storytelling, introducing dynamic narratives, plot choice, and direct embodiment of emotion.

While VR filmmaking is still maturing, facing challenges such as spatial restrictions and real-time user movement capture, promising innovations in camera layout, tracking devices, and rendering techniques are addressing these experiential issues. This paper explores technical solutions that enhance spatial freedom within VR films, with a particular focus on unlocking temporal freedom. We are driven by a shared interest in the narrative potential of VR filmmaking, that highlights the relationship between space, time, and the body. We propose a series of time-based ideas to enhance interactivity in VR films, moving beyond methods like voices, haptics, and controllers towards a more natural interaction with the story. This breakthrough is seen less as a technological advancement and more as a narrative and expressive breakthrough for VR filmmaking.

Inspired by the artworks of James Turrell and the ideas of psychologist James J. Gibson, we developed an experimental Omnidirectional Stereoscopic (ODS) interface aiming to provide users with an illusion of time. Turrell's use of the Ganzfeld effect, creating an infinite space through manipulated light, and Gibson's ambient optical array, influencing our visual perception of the environment, inspired our research. The experimental ODS interface in the VR context manipulates perceptual experiences, offering users a unique illusion of navigating through time as they explore the virtual environment.

3 RELATED WORKS

The rise of personal virtual reality (VR) devices and the emergence of VR film as a novel medium have generated significant interest in immersive content creation. VR films are primarily created using two methods: real-time computer graphics and photographically captured video employing 360 or Omnidirectional Stereo video cameras. A common challenge faced by photographically captured video is the inability to showcase content from different positions in space, undermining the potential of positional tracking and resulting in immersion-breaking experiences. A series of groundbreaking projects has tackled the challenges inherent in photographically captured VR content, aiming to enhance the creative and interactive potential of VR films. Notably, a Google AI team led by [6] delved into light field capture and rendering in VR, a significant stride in overcoming the limitations of positional tracking. Subsequent GPU pipeline optimizations by [3] further refined this technique. Other advancements include the implementation of recent developments in NeRF by [10] and Gaussian Splatting by [4], facilitating real-time rendering in VR. Novel view synthesis methods, such as 3D Reconstruction from Panoramas [5, 11], Planar Image Based Rendering [1], and Panoramic Image Based Rendering [8], have contributed to expanding the possibilities of VR content. Recent developments, like Free-Viewpoint Navigation of Indoor Scene with 360° Field of View by [9], showcase the ongoing progress in this domain. Additionally, the approach of selecting from a stored image array based on view and location has evolved over time, with early work on Viewpoint Dependent Imaging by [2] laying the foundation for Virtual Reality

development. More recent contributions by [7] have furthered the exploration of these techniques. Collectively, these advancements serve as the basis for our endeavor to implement such techniques into a standardized toolset for future VR filmmakers.

Collectively, these advancements not only overcome the limitations of photographically captured VR content but also set the stage for more creative and interactive VR film content.

4 TECHNICAL DETAILS

When filming, the exact spatial location of the camera is recorded for each frame. The recorded frames are stored in a spatial - temporal array. During playback, the system does not display frames sequentially in a traditional manner. Instead, as the viewer moves through the scene, the system displays frames based on the real-time tracked position of the headset. Additionally, we implemented a low latency random seek data structure that supported consistent sub-12ms frame pacing, an offline super-resolution pass to enhance effective PPD, and a real-time blending algorithm to increase the spatial density of available frames based on the research specifically on low-cost VR by [7]. A fascinating temporal effect emerges from the capture - playback technique. The experience is rather simple and intuitive and the flow of time is proportional to body movement. If the viewer moves forward, time advances. If the viewer moves backward, time rewinds. If the viewer stands very still, time slows till it freezes. Figure 1 illustrates this concept.

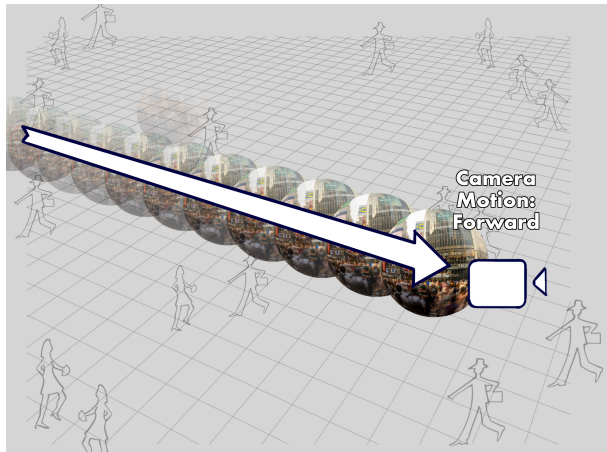
The viewer is essentially walking through space and time at once.

5 KEY CASE STUDY

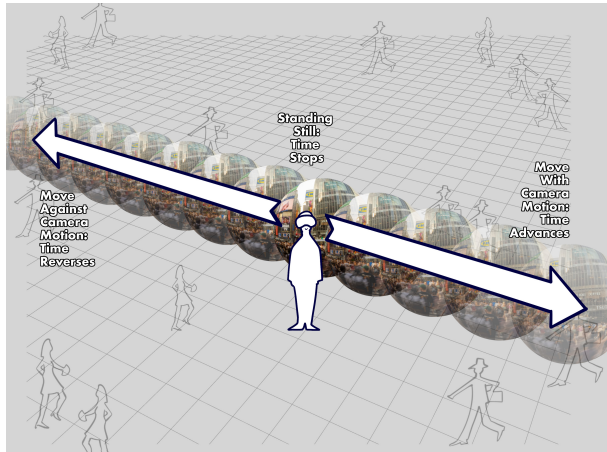
We set out to further experiment with the technique and to build a language and toolset for future filmmakers by creating a series of interactive experiences. First, to establish baseline perimeters, we experimented with the simplest scenario. We used videos in which the cameras traveled at constant speeds in straight lines, shot at the famous Shibuya crossing in Tokyo. We varied the speed of the camera movement and the framerate of the recording. With empirical controlled testing conducted with 20 users who had varying levels of experience with Virtual Reality, we identified two strongest factors contributing to nausea: inadequate frame density (skipping frames) and viewers deviating too far from the camera path.

Frame density is a function of camera speed (millimeters per second) divided by the video framerate (frames per second). Increasing the framerate requires more sophisticated (and expensive) camera hardware and degrades image quality. We also would like to keep the camera speed flexible as a creative expression that can be controlled. We found that any frame density less than 5 millimeters per frame will cause significant nausea. Under such density, offline rendering and storing each frame for a camera path that can be as long as 20+ meters puts too much strain on storage. The optimal solution became real-time blending and generation of novel views [7]. We were able to implement their research and achieve a consistent sub-12ms frame pacing.

Pacing Time further explored the narrative potential in a circular, infinite loop. It was installed as an immersive exhibition located in a 1500 square foot black box gallery. The exhibition had a physical



(a) Camera linear motion during filming creating a series of omnidirectional stereoscopic views



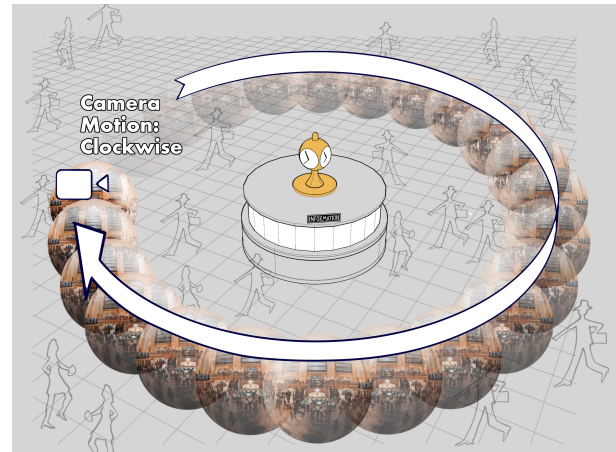
(b) Viewer linear motion plays back the corresponding views, creating the illusion of walking through time.

Figure 1: Illustration of the temporal effect achieved through the capture-playback technique.

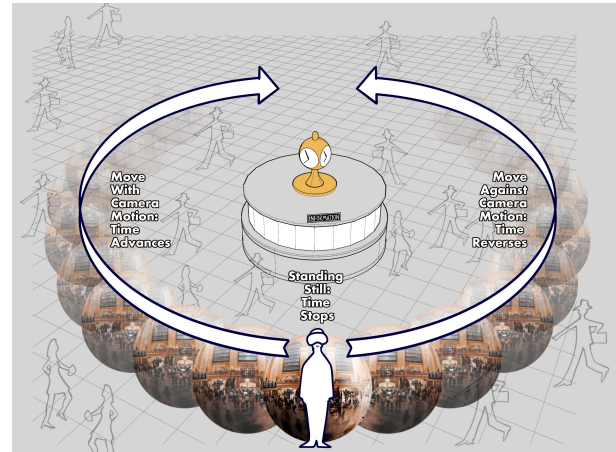
component and an immersive component that worked together to demonstrate the relationship of the body to time in the ODS.

- Locomotion redirection techniques were implemented to represent the experience of a large space in a confined gallery.
- A physical component was included that served both technical and storytelling multiple functions. It helped anchor the viewers to the camera path, aided locomotion redirection, and served as a narrative component, a space-time fabric.

The physical component was a large thirty-foot diameter cylindrical shaped paper ring hanging from the ceiling. The paper ring was used to guide users through the immersive film, they would gently walk forward (or backward) while touching the edge of the ring as they wore the headset. This presented an interesting offset between physical and virtual space as in the virtual space, users thought they were walking in a straight line. The remapping of infinite distance in virtual space, to a circle in physical space, was



(a) Camera circular clockwise motion during filming creating a series of omnidirectional stereoscopic views



(b) Viewer circular motion plays back the corresponding views, creating the illusion of walking through time.

Figure 2: Illustration of the temporal effect achieved through the circular capture-playback technique.

inspired by James Turrell's Ganzfeld effect, however we were doing this with continuous movement and no boundaries.

6 CONCLUSION

Our exploration of locomotion's impact on narrative construction in Virtual Reality (VR) filmmaking, as exemplified by the innovative system presented in this paper, marks a significant opportunity for VR filmmaking. Pacing Time, our immersive installation and primary case study, challenges conventional locomotion methods ODS unveiling a novel connection between memory, the human body, and virtual space.

Looking toward the future of filmmaking, the convergence of our time-based techniques with emerging technologies and standardized tools for VR filmmakers holds great promise. The technical details of our system, from spatial recording of the camera's exact location for each frame to the implementation of low-latency random

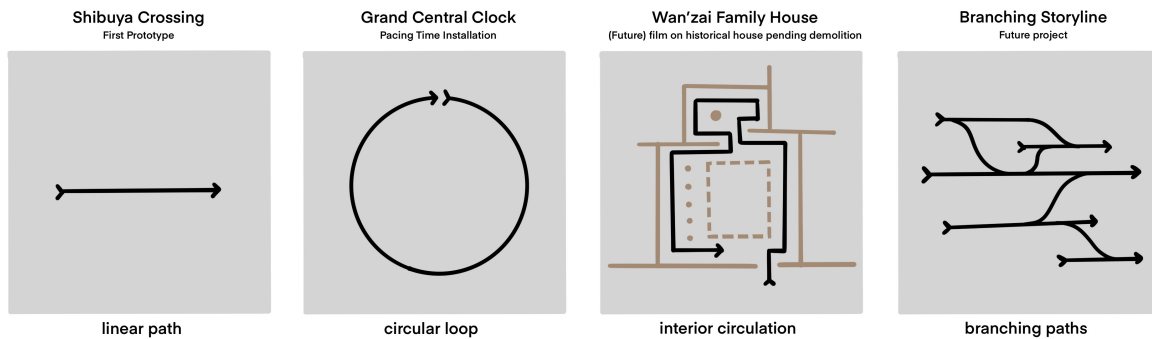


Figure 3: Other potential topologies for camera and viewer movement paths

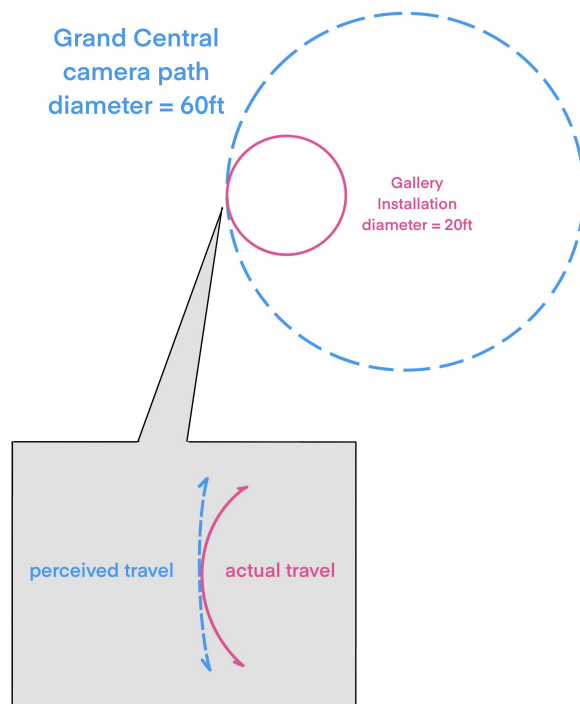


Figure 4: Locomotion redirection.

seek data structures and real-time blending algorithms, underscore the potential for a new era in VR storytelling. The temporal effect, where the viewer walks through space and time simultaneously, creates an immersive experience with a proportional flow of time to body movement. We hope that this paper contributes to the ongoing dialogue, sparking curiosity, innovation, and a renewed sense of artistic expression within the medium.

REFERENCES

- [1] Gaurav Chaurasia, Sylvain Duchene, Olga Sorkine-Hornung, and George Dretakis. 2013. Depth synthesis and local warps for plausible image-based navigation. *ACM Transactions on Graphics* 32, 3 (2013), 30. <https://doi.org/10.1145/2487228>.



Figure 5: Physical + digital installation



Figure 6: Physical + digital installation

2487238

- [2] Scott Fisher. 1983. Viewpoint Dependent Imaging: An Interactive Stereoscopic Display. In *Processing and Display of Three-Dimensional Data*, Vol. 0367. SPIE. <https://doi.org/10.1117/12.934299>
- [3] S. Hajisharif, E. Miandji, P. Larsson, K. Tran, and J. Unger. 2019. Light Field Video Compression and Real Time Rendering. *Computer Graphics Forum* 38 (2019). <https://doi.org/10.1111/cgf.13835>

- [4] Bernhard Kerbl, Georgios Kopanas, Thomas Leimkuehler, and George Drettakis. 2023. 3D Gaussian Splatting for Real-Time Radiance Field Rendering. *ACM Transactions on Graphics* 42, 4 (2023), 139. <https://doi.org/10.1145/3592433>
- [5] P.K. Lai, S. Xie, J. Lang, and R. Laganière. 2019. Real-Time Panoramic Depth Maps from Omni-directional Stereo Images for 6 DoF Videos in Virtual Reality. In *IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*. 405–412.
- [6] Ryan S. Overbeck, Daniel Erickson, Daniel Evangelakos, Matt Pharr, and Paul Debevec. 2018. A System for Acquiring, Processing, and Rendering Panoramic Light Field Stills for Virtual Reality. *ACM Transactions on Graphics* 37, 6 (2018), 197. <https://doi.org/10.1145/3272127.3275031>
- [7] Matias Selzer, M. Ganuza, and Silvia Castro. 2021. High Visual-Quality Scenes in Low-Cost Virtual Reality With Collisions and Irregular Surfaces. *IEEE Computer Graphics and Applications* (2021). <https://doi.org/10.1109/MCG.2021.3068037>
- [8] M. Uyttendaele, A. Criminisi, S. B. Kang, S. Winder, R. Szeliski, and R. Hartley. 2004. Image-based interactive exploration of real-world environments. *IEEE Computer Graphics and Applications* 24, 3 (2004), 52–63. <https://doi.org/10.1109/MCG.2004.1297011>
- [9] H. Xu, Q. Zhao, Y. Ma, S. Wang, C. Yan, and F. Dai. 2023. Free-Viewpoint Navigation of Indoor Scene with 360° Field of View. *Electronics* 12, 8 (2023), 1954. <https://doi.org/10.3390/electronics12081954>
- [10] Linning Xu, Vasu Agrawal, William Laney, Tony Garcia, Aayush Bansal, Changil Kim, Samuel Rota Bulò, Lorenzo Porzi, Peter Kotschieder, Aljaž Božić, Dahua Lin, Michael Zollhöfer, and Christian Richardt. 2023. VR-NeRF: High-Fidelity Virtualized Walkable Spaces. In *SIGGRAPH Asia 2023 Conference Papers*. ACM, 1–12. <https://doi.org/10.1145/3610548.3618139>
- [11] N. Zioulis, A. Karakottas, D. Zarpalas, and P. Daras. 2018. OmniDepth: Dense Depth Estimation for Indoors Spherical Panoramas. In *European Conference on Computer Vision*. 453–471.